

GENERAL GUIDELINES

SCHEDULE OF THE SYMPOSIUM

Date: 5th November

The symposium will be conducted in two parts, each allotted a dedicated time slot. The IT segment will take place first, followed by the Business Product Pitch event.

Part 1: IT Segment

Top 6 Selected essays per category of the research paper will be presented by their respective authors.

Time	Activity
9:00 – 9:15	• Welcome by Mrs Kar/ Head Girl • Opening Keynote address by CSIR (Indian Institute of Chemical Biology)
9:15 – 10:15	IT Interdisciplinary Paper <i>(5 minutes presentation + 5 minutes audience Q&A per paper)</i>
10:15 – 11:15	IT Technical Paper <i>(5 minutes presentation + 5 minutes audience Q&A per paper)</i>
11:15 - 11:30	Ending Keynote address by RCCIIT
11:30 – 12:00	Break

Part 2: Business Product Pitch Event

Time	Activity
12:00 - 12:15	• Opening address by Principal/ Deputy Head Girl • Address by Moderator

12:15 – 2:45	Business Product Pitch Event <i>(18–20 minutes allotted per school, inclusive of presentation and Q&A by the audience and judges)</i> <i>(The judges will be addressing the audience about the presentations and pitches after each school goes up till the 6th performance)</i>
2:45– 3:00	Closing Ceremony

Note: Between each presentation, the judges will share brief remarks on the symposium and the products presented so far.

DETAILS ABOUT THE BUSINESS PRODUCT PITCHING EVENT:

Timeline of the Business Product Pitching event

1. The proposal will be sent to 20 schools.
2. Registration for the event will be conducted on a First Come, First Served basis once the proposals are sent out.
3. **Only** the first 12 schools to register will be given the opportunity to participate in the event.
4. All 12 participating schools must submit a brief but comprehensive presentation with substantial information about the product. This can be accompanied by a video/pictures of the prototype (optional) of their product to the host school one month prior to the symposium, i.e., by the **first week of October**.
5. The submitted presentations will undergo a preliminary screening, 6 schools will be shortlisted based on how well their presentations meet the required criteria and quality expected at the symposium.
6. By **mid-October**, the host school will notify the 6 qualifying schools of their selection.
7. The 6 qualifying schools will then present their final presentation, along with their final product, on the day of the symposium.

DETAILS ABOUT THE IT RESEARCH PAPER EVENT:

Timeline of the IT Research Paper event

1. The proposal will be sent to 20 schools.
2. All interested participants from the participating schools must submit their papers to the host school one month prior to the symposium, i.e., by the **first week of October**.
3. The submitted papers will undergo a preliminary screening. 12 papers (6 for technical and 6 for interdisciplinary) will be selected based on how well they meet the required criteria and quality expected at the symposium.
4. By **mid-October**, the host school will notify the qualifying participants of their selection.
5. The 12 qualifying participants will then present their final paper on the day of the symposium.

IT (TECHNICAL)

IT Research Paper Proposal - Technical

Event Overview

1. The symposium will invite **senior school** students (Grades 11 and 12) to submit and present a technical research-based thesis paper.
2. It will be addressing a real-world problem within the domain of **Cybersecurity and Digital Infrastructure**.
3. Preferred format - **IEEE format** (as attached)
4. **RCCIIT** is proposed as the named academic partner and certifying authority for this track. They can help in lending subject-matter credibility and a formal evaluation panel to the students' work.

About the papers:

- Should be **data-driven** and analytical, rather than descriptive
 - Should include problem identification → research → solution → evaluation
 - Should be supported with case studies, **primary data** and statistics
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DETAILS OF THE THESIS

Suggested Approaches:

- **Threat and Risk Analysis:**

Examine a real-world cyberattack or security incident. Evaluate its causes, impact and what it reveals about gaps in preparedness or response.

- **Case Study of Sectoral Impact:**

Analyse how a specific cyber threat or vulnerability class affects a particular sector or user group (schools, small businesses, healthcare, public services). Identify structural gaps in awareness, process or technical controls.

- **Solution-Oriented Research Paper:**

Identify a specific cybersecurity or data-privacy challenge. Propose a credible, evidence-backed, feasible technical or procedural solution. It should be grounded in recognised frameworks or standards (e.g., NIST, ISO 27001) where relevant.

Structural Template:

1. Executive Summary
2. Introduction & Issue Framing
3. Literature Review (It shows what is already known, points out gaps or flaws in existing research and sets a strong foundation for your own new research)
4. Methodology
5. Findings & Analysis (Data-Driven)
6. Current Practice or Policy Response (only if relevant)
7. Proposed Solutions with Feasibility Assessment
8. Limitations & Counterarguments
9. Conclusion with Predicted Outcomes of the Research
10. References & Appendices

Suggested Questions:

1. How well do different types of multi-factor authentication (push notifications, biometrics, etc.) resist common attacks like SIM-swapping or session hijacking?
2. How well do current phishing-detection tools (email filters, URL checks, ML-based classifiers) catch attacks designed to fool humans, not just machines?
3. Focussing on any one specific type of IoT device (e.g., smart home cameras), how do weak encryption or insecure APIs leave it open to attack?
4. Choose a specific popular app or platform. Do their data-collection practices actually match their stated privacy policy?
5. How well do Indian institutions' technical security measures (network segmentation, access controls, encryption) meet recognised cybersecurity standards?

Judging Criteria

1. Understanding of IT & AI/ML Concepts
 2. Breadth of Threat/Topic Analysis
 3. Real-World Relevance
 4. Critical Thinking & Argumentation
 5. Feasibility of Solutions & Impact Potential
 6. Use of Data & Evidence
 7. Presentation & Persuasiveness
 8. Innovation and Creativity
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IT (INTERDISCIPLINARY)

IT Research Paper Proposal - Interdisciplinary

Event Overview

1. The symposium will invite **senior school** students (Grades 11 and 12) to submit and present an interdisciplinary research-based thesis paper.
2. It will be addressing a real-world problem within the domain of **Cybersecurity and Digital Infrastructure** while including an element from another domain of their choice. The overarching theme will be **Technology and Society**.
3. Preferred format - **IEEE format** (as attached)
4. **CSIR (Indian Institute of Chemical Biology)** is proposed as the named academic partner and certifying authority for this track. They can help in lending subject-matter credibility and a formal evaluation panel to the students' work.

About the papers:

- Should build a **clear, defensible thesis** and engage substantively with existing scholarship in your chosen discipline, not merely offer opinion.
 - Should be **argument and evidence-driven** rather than data/methodology-driven: you're expected to engage seriously with disciplinary theory and existing scholarship and to build a reasoned position, rather than run an experiment or survey.
 - Should keep the **technology dimension central** to your question. This keeps the track distinct from a general essay competition and gives evaluators a coherent area to assess.
 - May use **primary evidence** where it strengthens your argument (interviews, case studies of specific policies or events, or small opinion surveys) as support, rather than as your core method.
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DETAILS OF THE THESIS

Suggested Approaches:

- **Comparative Case Analysis:**

Examine two contrasting cases to draw out a broader argument about the chosen theme. Use the comparison to test or refine a claim.

- **Theoretical Application:**

Apply an established theory or framework from the chosen discipline (philosophical account of agency, a theory of state power, a sociological theory of trust, etc) to a contemporary AI or cybersecurity issue. Assess how well it holds up.

- **Policy or Ethical Critique:**

Critically evaluate an existing law, policy or institutional stance related to AI, data privacy or cybersecurity. Identify its underlying assumptions and propose a reasoned alternative grounded in the discipline's theory.

Structural Template:

1. Abstract
2. Introduction & Thesis Statement
3. Disciplinary Framework / Theoretical Grounding
4. Argument & Analysis (with supporting evidence or case studies)
5. Counterarguments & Their Rebuttal
6. Broader Implications
7. Conclusion
8. References & Appendices (if applicable)

Suggested Questions:

1. With artificial intelligence increasingly taking over decisions in surveillance and digital infrastructure, does anything remain uniquely human about human thought, agency or consciousness?
2. How is public trust in institutions (media, government, banking, etc) changing as surveillance, data collection and algorithmic decision-making expand into everyday life?

3. What can earlier periods of rapid technological disruption (the Industrial Revolution, the early Internet, etc) teach us about how digital infrastructure and governance adapt, or fail to adapt, to the AI era?
4. 'Digital Divide 2.0' - Does the rapid development of digital infrastructure and AI lessen social inequality or does it lead to new kinds of technical exclusion?
5. What obligations should technology platforms have, and who should enforce them, to prevent deepfakes, misinformation and AI-generated content from undermining democratic decision-making?

Judging Criteria

1. Disciplinary Understanding
 2. Real-World Relevance
 3. Originality & Argumentation
 4. Use of Data & Evidence
 5. Engagement with Counterarguments
 6. Clarity & Structure
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BUSINESS

ACADEMIC SYMPOSIUM BUSINESS PROPOSAL

A Product Pitch for Student Entrepreneurs

From Idea to Enterprise: A Research-Backed Pitch by Student Founders

1. Introduction & Rationale

The Business Symposium is an academic event for students of Classes 11-12, designed to go beyond conventional project displays and conventional shark tank competitions. It invites students to develop a single business idea into a fully reasoned, evidence-backed venture proposal. Students may either build on an idea they have previously presented at a school-level competition, strengthening it with deeper research, data and strategy.

2. Objectives

- To cultivate entrepreneurial thinking, financial literacy, and evidence-based reasoning among students.
 - To teach students how to turn a raw idea into a structured business case using real analytical frameworks (SWOT, USP, scalability, and CSR).
 - To reward depth of research and iteration by allowing improvement of previously presented ideas, rather than demanding novelty alone.
 - To connect students with a panel with those who are experts in the fields of presentation given below in order to determine the viability and credibility of the product innovation.
 - To embed social responsibility and sustainability thinking into student enterprise from an early stage.
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3. Format & Structure

3.1 Team Composition

Each team will comprise 4 students, who will together present different aspects of the product including its financing, usability, market value, and research.

3.2 Categories of Ideas

Teams may present either:

- **Elevated Idea:** A concept previously presented at a school-level event (exhibition, fest, or competition), now revisited with fresh research, financial modelling, and a viability study.
- **New Venture:** An entirely original idea developed from scratch for the Symposium. *Note: Both categories are judged equally; there is no separate scoring track for either.*

3.4 Themes for the product:

- **Circular Economy:** A business model that minimizes waste by keeping products and materials in use through reuse, repair, refurbishment, and recycling, instead of the traditional "take-make-dispose" approach.
- **Gig Economy:** A labour model based on short-term, flexible, or on-demand work rather than permanent employment, coordinated mainly through digital platforms.

3.5 Important Notes

- The prototype must be safe and functional
- Working models are strongly encouraged
- Tech solutions: show working code or a physical prototype
- Non-tech solutions: mockups, storyboards, or visual walkthroughs are acceptable

4. Stages of Competition

Each team will present for **7 minutes**, that must not be exceeded. The presentation

should showcase the product in its final form, rather than a prototype. This will be followed by **5 minutes** of panel cross-questioning, after which the floor will open to audience questions for deeper discussion.

Teams must conduct **primary research** (a short survey, 10–15 interviews, or a field test) and **secondary research** (industry reports, government data, news) to support the product's viability.

5. Mandatory Elements of Every Pitch

To maintain a consistent standard, every proposal and live pitch must include:

- **Problem Statement:** The specific problem being solved, with evidence that it exists.
- **SWOT Analysis:** An honest evaluation of Strengths, Weaknesses, Opportunities, and Threats.
- **Unique Selling Proposition (USP):** What sets the idea apart from existing alternatives.
- **Target Market & Customer Segment:** Who the product/service is for, with basic market sizing (TAM-SAM-SOM Analysis, if possible).
- **Business & Revenue Model:** How the venture earns money ie. pricing, channels, and cost structure.
- **Scalability Plan:** How the idea can grow beyond its pilot stage including geographically, digitally, or across product lines.
- **Viability & Financial Reasoning:** Approximate cost estimates, break-even reasoning, and funding requirements, even in simplified form.
- **Competitive Analysis:** At least two comparable products/services and how the idea outperforms them.
- **CSR / Sustainability Angle:** The social, environmental, or community benefit built into the venture from the start, not added afterward.
- **Risk Analysis & Mitigation:** Key risks to the idea and how the team plans to manage them.
- **Prototype / Proof of Concept:** A model, sample, app mock-up, working demo, or detailed visual mock-up that makes the idea tangible.

6. Judging Criteria

All the given criterias have equal weightage and carry 10 marks each

- Problem & Research Evidence
- SWOT & Risk Analysis
- USP & Competitive Edge
- Target Market & Sizing
- Business & Revenue Model
- Financial Viability
- Scalability
- CSR / Sustainability & Theme Fit
- Prototype / Proof of Concept
- Pitch Delivery & Cross-Questioning

7. Conclusion

The Business aspect of the Academic Symposium is designed to be a one day event, with learning and impact that students can takeaway for a lifetime; it is intended as a culture-building exercise that trains students to think like founders, reason like analysts, and present like professionals.
